

PC Upper Computer Software User Manual

Electronic Loads MP LOAD Series



Part Number: MP710259 US

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1. Preface

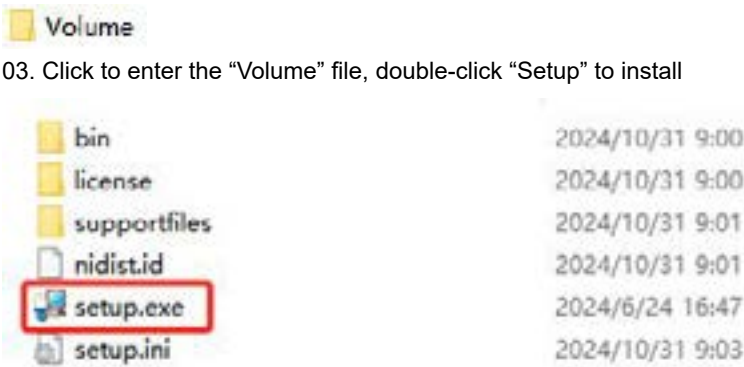
This software is suitable for remote control of MP LOAD Series power supplies. The above series also include multiple models, and there are differences in various functions among different models, such as some models without external control functions or some parameters not open for setting. The operating logic is basically the same. This document provides remote control operation explanations for the most comprehensive models.

2. Software installation

The software installation requires Windows 7 or higher operation systems. The following are the installation steps.

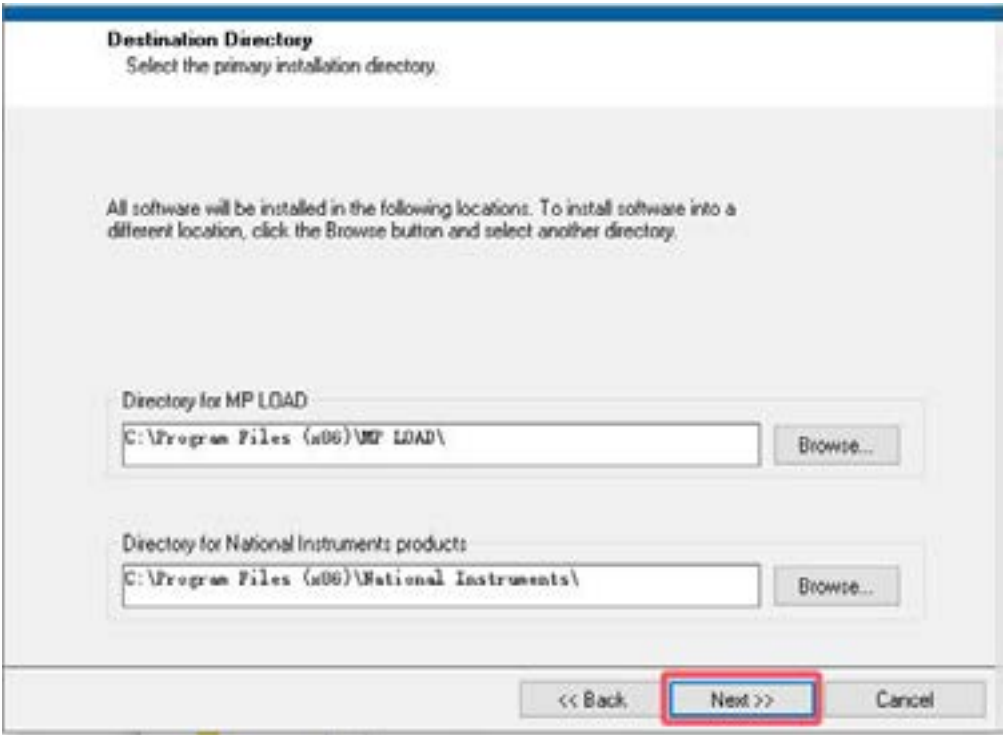
01. After obtaining the remote control software compressed package, decompress it.

02. After decompression, the following file will be obtained

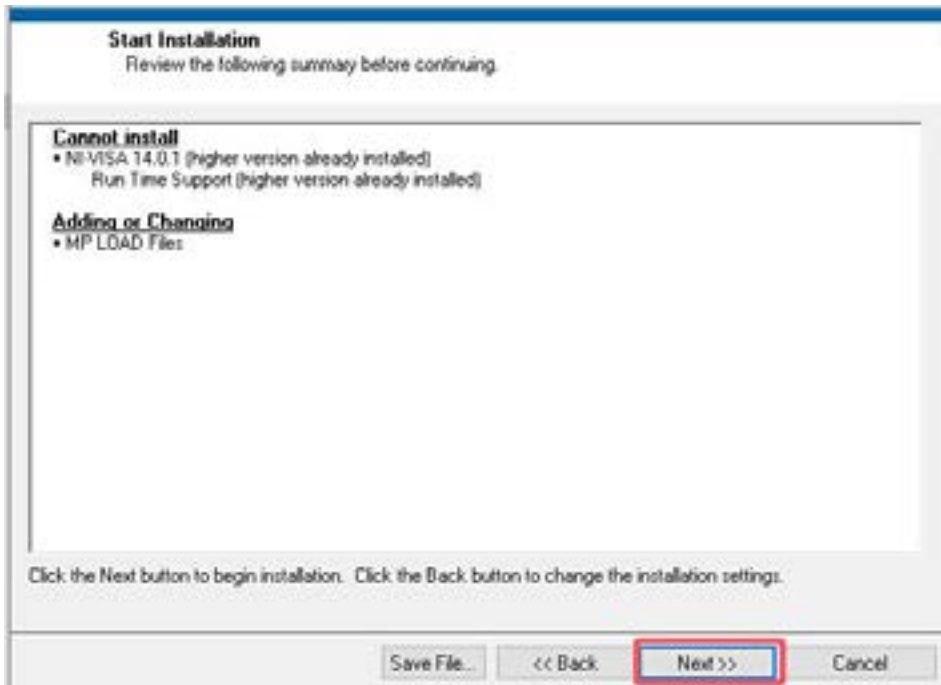


03. Click to enter the “Volume” file, double-click “Setup” to install

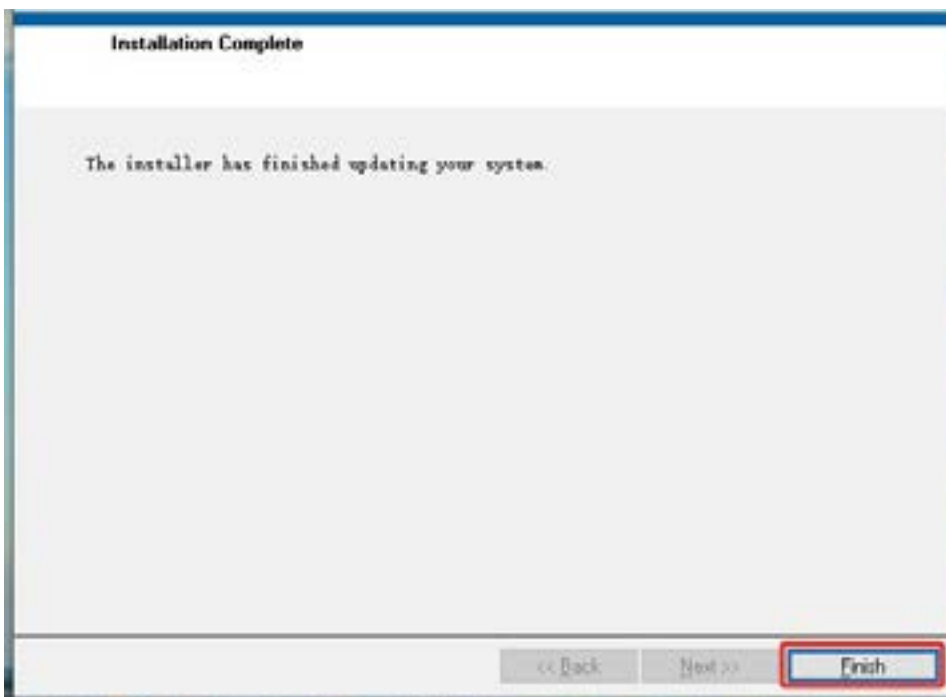
04. After the initialization of the installation package is completed, you can change the installation path according to your needs, and then click “Next”



05. Continue clicking 'Next'



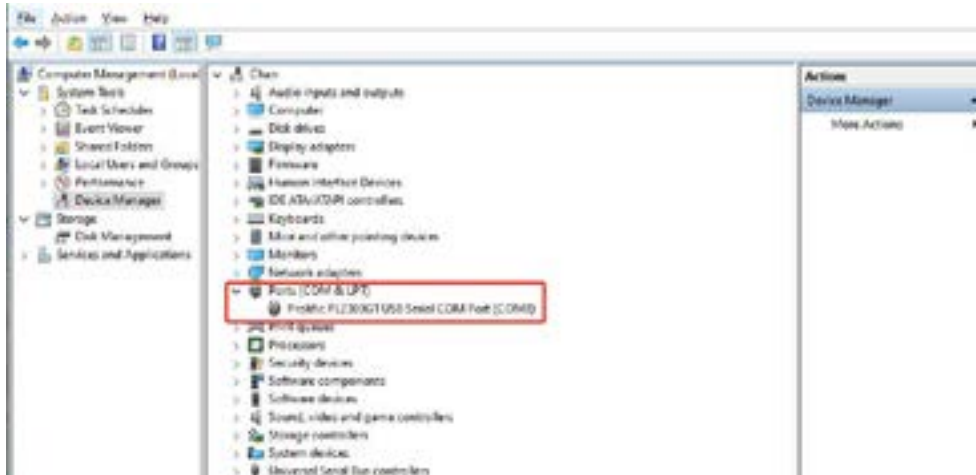
06. Wait for a moment, complete the installation by clicking "Finish"; and then find the installed software and proceed with the operation.



3. Software startup and interface operation instructions

3.1 Selection of Communication Ports

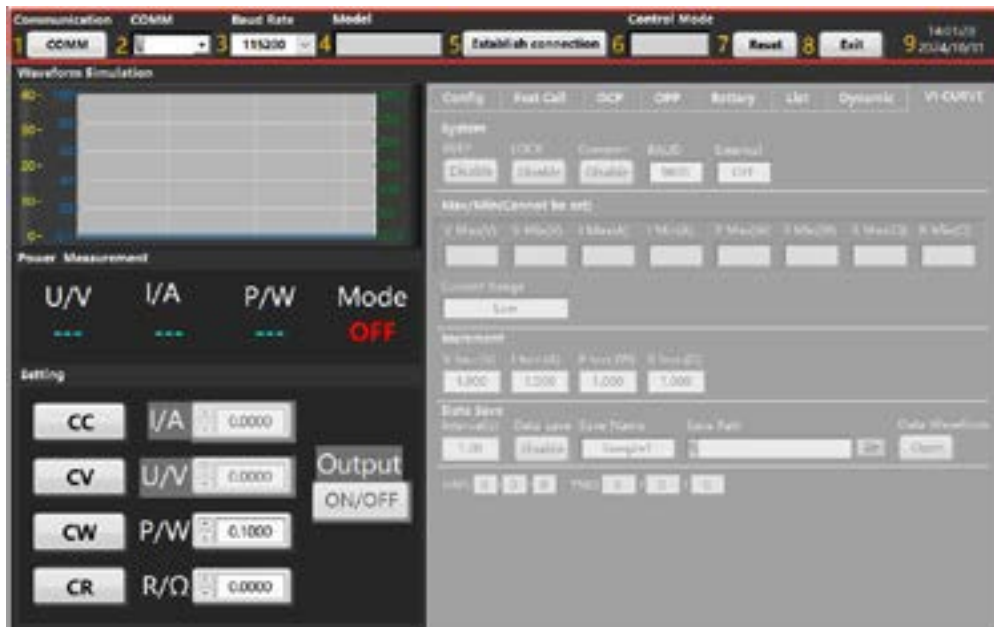
The communication methods include RS232, USB, and LAN. At first, connect the PC and the power supply by using the provided or corresponding communication cable. After inserting the corresponding communication cable, select the corresponding communication method and configure the corresponding communication settings to perform remote control. If you find that communication is not working properly, you can open the Device Manager on my computer and check if the corresponding computer communication port is working properly.



USB serial device abnormalities may occur in the Windows 7 system, and exclamation marks may appear on the above interface. In this case, it is necessary to obtain the corresponding USB driver from the supplier for installation.

3.2 Interface Operation Instructions

3.2.1 Communication & State



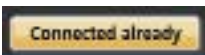
01. Figure number 1, select communication methods COMM and LAN. As the USB communication cable will transform into a COMM port after insertion, USB here belongs to COMM port communication. And select the corresponding COMM port number. If choosing RS232 communication cable, select COMM here; if selecting USB communication cable, select COMM here; if selecting LAN, click here to switch to LAN mode. When selecting LAN mode, you can configure the IP and network ports. The above parameters need to be consistent with the device panel display, as shown in the following figure:



02. Number 2 in the diagram, select the COMM port. In the device management section, check the port number for connecting the computer to the device, and click the drop-down box to select the corresponding COMM port.

03. Figure 3, Baud Rate - baud rate setting, it requires the corresponding baud rate to be configured on the device panel. Select the corresponding baud rate here.

04. After the communication configuration is correct, click here to establish a communication connection. Once the connection is successful, this area will light up, indicating that the communication has been connected. The software here takes 1-2 seconds to initialize and lock the device. And some parameters will be set to default values, please pay attention to parameter changes.



05. After successful connection, icon number 4 will display the recognized model, and icon number 5 will display the current control mode, which is divided into remote control and external control.

06. Click Reset to reset the configuration parameters.

07. Click Exit to exit the system.

3.2.2 Power Measurement



Real time display of data waveform, read back voltage, current, power, and output mode.

3.2.3 Settings



The setting area can set the output modes CC/CV/CW/CR, the corresponding output values and control outputs of the modes.

3.2.4 Configuration



3.2.4.1 System



BEEP: the buzzer is turned on and off, and the button will light up when turned on. LOCK: press the button to lock and unlock, and the button will light up when turned on. Compen: remote compensation on and off, the button will light up when turned on. BAUD: baud rate setting External: the selection of external control can be done in Trig mode or Switch mode.

3.2.4.2 Max/Min



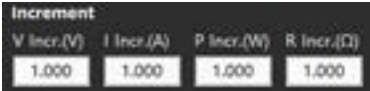
Display the maximum and minimum voltage, maximum and minimum current, maximum and minimum power, and maximum and minimum resistance values of the current device.

3.2.4.3 Current Range

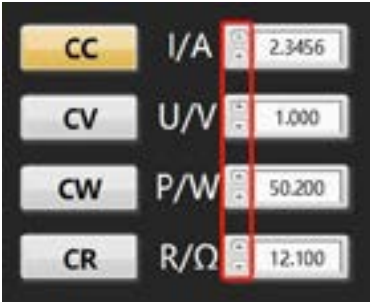


Click to switch the current level, and select High or Low.

3.2.4.4 Increment



Incremental setting: after setting the target value, the voltage, current, power, and resistance of settings will increase incrementally, which is used for fine-tuning operations as shown in the following figure:



3.2.4.5 Data Save

01. The use of Data Save: Interval (s) data - recording time interval;
Save Name saved document name;
Save Path - data saving path;

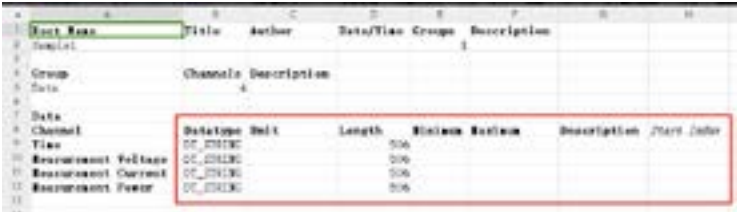
defaults to saving in the root directory of drive D. If there is no drive D, it will be saved to the root directory of drive C.



02. After configuring the above parameters, click , the data begins to be recorded, and the corresponding parameters can not be changed. If you need to modify them, click again to turn off the data recording function. If Save Name has not been changed, the data will continue to be saved in the previous file. If it has been changed, a new data file will be created.

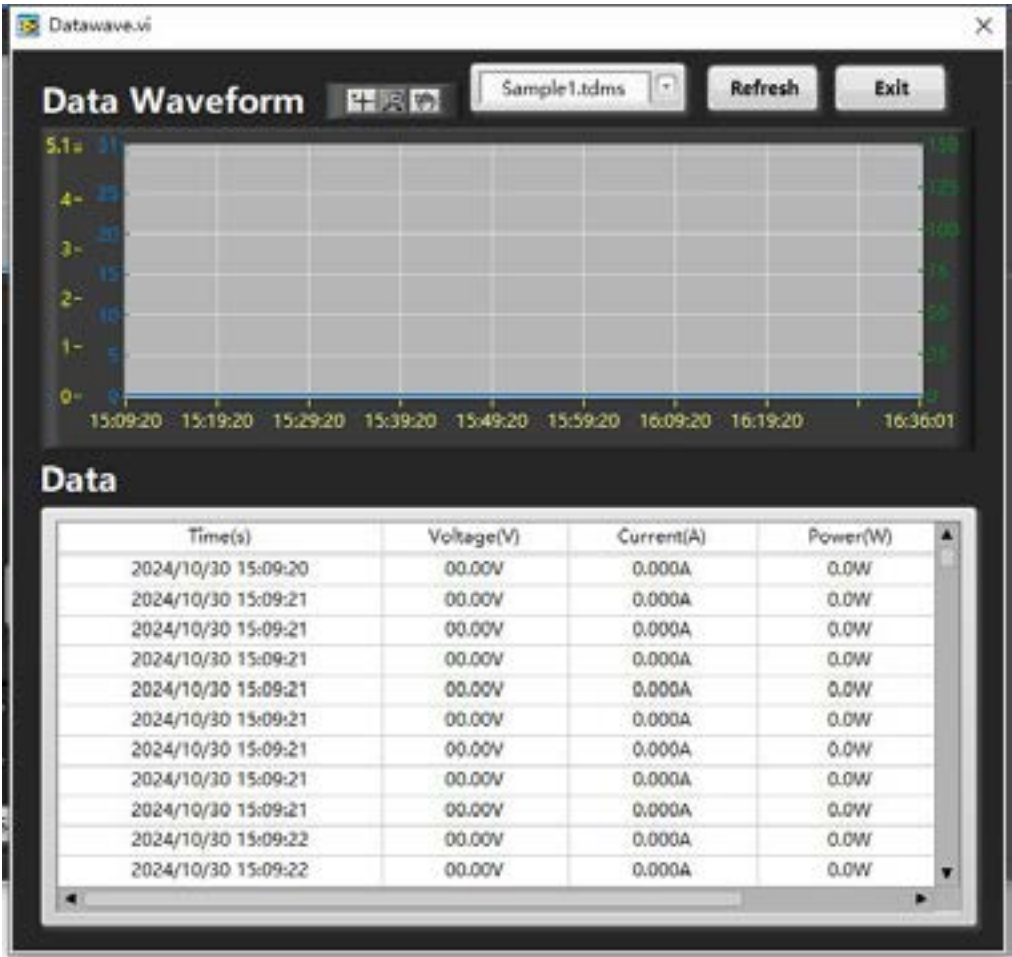


Data file description: users can find the data file based on the displayed path here. The saved file format is. tdms file, which can be opened by Excel. The file will display the data quantity and real-time data. And switch the table name in the lower left corner.



#	A	B	C	D
1	Time	Measurement Voltage	Measurement Current	Measurement Power
2	2024/10/21 17:05:19	20.00V	0.000A	0.0W
3	2024/10/21 17:05:19	20.00V	0.000A	0.0W
4	2024/10/21 17:05:20	20.00V	0.000A	0.0W
5	2024/10/21 17:05:21	20.00V	0.000A	0.0W
6	2024/10/21 17:05:22	20.00V	0.000A	0.0W
7	2024/10/21 17:05:23	20.00V	0.000A	0.0W
8	2024/10/21 17:05:24	20.00V	0.000A	0.0W
9	2024/10/21 17:05:25	20.00V	0.000A	0.0W
10	2024/10/21 17:05:26	20.00V	0.000A	0.0W
11	2024/10/21 17:05:27	20.00V	0.000A	0.0W
12	2024/10/21 17:05:28	20.00V	0.000A	0.0W
13	2024/10/21 17:05:29	20.00V	0.000A	0.0W
14	2024/10/21 17:05:30	20.00V	0.000A	0.0W
15	2024/10/21 17:05:31	20.00V	0.000A	0.0W
16	2024/10/21 17:05:32	20.00V	0.000A	0.0W
17	2024/10/21 17:05:33	20.00V	0.000A	0.0W
18	2024/10/21 17:05:34	20.00V	0.000A	0.0W
19	2024/10/21 17:05:35	20.00V	0.000A	0.0W
20	2024/10/21 17:05:36	07.60V	3.553A	26.9W
21	2024/10/21 17:05:37	02.22V	5.100A	11.3W
22	2024/10/21 17:05:38	02.22V	5.100A	11.3W
23	2024/10/21 17:05:39	02.22V	5.100A	11.3W
24	2024/10/21 17:05:40	02.22V	5.100A	11.3W
25	2024/10/21 17:05:41	02.22V	5.100A	11.3W
26	2024/10/21 17:05:42	02.22V	5.100A	11.3W
27	2024/10/21 17:05:43	02.22V	5.100A	11.3W
28	2024/10/21 17:05:44	02.22V	5.100A	11.3W
29	2024/10/22 10:09:51	22.94V	0.000A	0.0W
30	2024/10/22 10:09:52	22.94V	0.000A	0.0W
31	2024/10/22 10:09:53	22.94V	0.000A	0.0W
32	2024/10/22 10:09:54	22.94V	0.000A	0.0W
33	2024/10/22 10:09:55	22.94V	0.000A	0.0W
34	2024/10/22 10:09:56	22.94V	0.000A	0.0W
35	2024/10/22 10:09:57	22.94V	0.000A	0.0W
36	2024/10/22 10:09:58	22.94V	0.000A	0.0W
37	2024/10/22 10:09:59	22.94V	0.000A	0.0W
38	2024/10/22 10:10:00	22.94V	0.000A	0.0W
39	2024/10/22 10:10:01	22.94V	0.000A	0.0W
40	2024/10/22 10:10:02	22.94V	0.000A	0.0W
41	2024/10/22 10:10:03	22.94V	0.000A	0.0W
42	2024/10/22 10:10:04	22.94V	0.000A	0.0W
43	2024/10/22 10:10:05	22.94V	0.000A	0.0W
44	2024/10/22 10:10:06	22.94V	0.000A	0.0W
45	2024/10/22 10:10:07	22.94V	0.000A	0.0W
46	2024/10/22 10:10:08	22.94V	0.000A	0.0W
47	2024/10/22 10:10:09	22.94V	0.000A	0.0W
48	2024/10/22 10:10:10	22.94V	0.000A	0.0W
49	2024/10/22 10:10:11	22.94V	0.000A	0.0W
50	2024/10/22 10:10:12	22.94V	0.000A	0.0W
51	2024/10/22 10:10:13	22.94V	0.000A	0.0W
52	2024/10/22 10:10:14	22.94V	0.000A	0.0W
53	2024/10/22 10:10:15	22.94V	0.000A	0.0W
54	2024/10/22 10:10:16	22.94V	0.000A	0.0W
55	2024/10/22 10:10:17	22.94V	0.000A	0.0W
56	2024/10/22 10:10:18	22.94V	0.000A	0.0W
57	2024/10/22 10:10:19	22.94V	0.000A	0.0W
58	2024/10/22 10:10:20	22.94V	0.000A	0.0W
59	2024/10/22 10:10:21	22.94V	0.000A	0.0W
60	2024/10/22 10:10:22	22.94V	0.000A	0.0W
61	2024/10/22 10:10:23	22.94V	0.000A	0.0W
62	2024/10/22 10:10:24	22.94V	0.000A	0.0W
63	2024/10/22 10:10:25	22.94V	0.000A	0.0W
Sample1 (root) Data +				

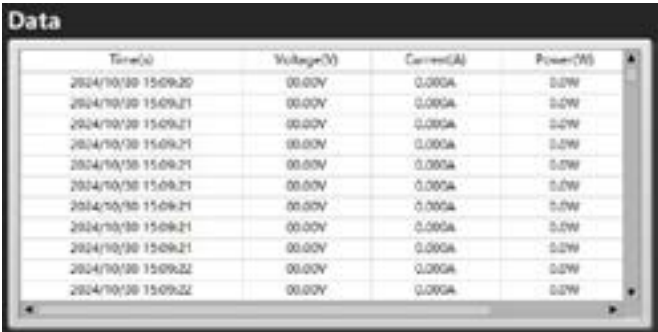
03. Data Waveform - After clicking, it will enter the data waveform interface.



Click the drop-down menu to select the file to display.



Data refresh button - click to refresh the latest data during the recording process.



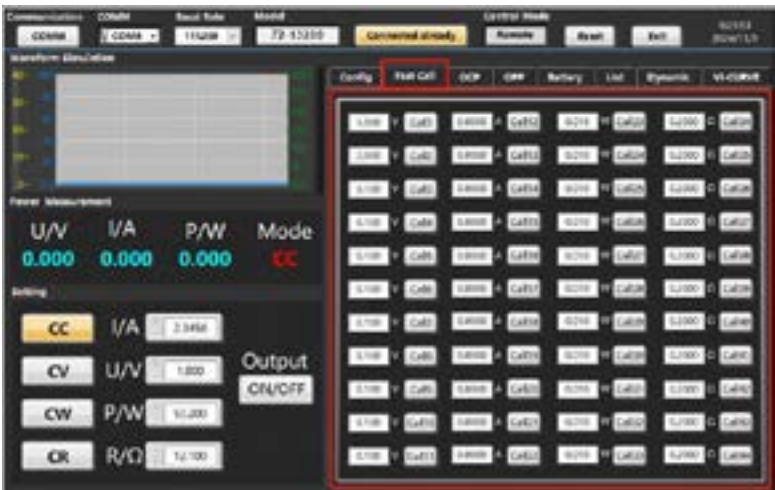
Display all the data in the file and scroll the wheel to view it.

3.2.4.5 Time



HMS: Set the device's hours, minutes, and seconds;
YMD: Set the device's year, month, and date.

3.2.5 Fast Call



A total of 44 sets of parameters are available for users to call at any time, and will be automatically saved to the Datasave path when the software exits.

3.2.6 OCP/OPP



After entering OCP mode, the Mode of Power Measurement will change to the red font OCP mode, indicating that it has entered OCP mode but is not running. Then, it will automatically read the parameters stored in the device, totaling 7 sets. After setting the corresponding parameters, select the set of parameters to be executed in the OCP CALL NO. option box, and finally click ON/OFF on the left to run. At this time, the Mode OCP font will turn green, indicating that it is running. The same applies to OPP mode operation.



3.2.7 Battery



After entering battery mode, the Mode of Power Measurement will change to BATT mode in red font, indicating that it has entered BATT mode but is not running. Then, it will automatically read the parameters stored in the device, totaling 7 sets. After setting the corresponding parameters, select the set of parameters to be executed in the BATT CALL NO. option box, and finally click ON/OFF on the left to run. At this time, the Mode BATT font will turn green, indicating that it is running.

Parameter description:

Capacity (AH): After running, it displays the accumulated ampere hours of this run.

Time (S): After running, display the accumulated time of this run.

Last run stop event: Record the cut-off event after the last run, such as the end of time, reaching the ampere hour parameter, or reaching the cut-off voltage.

3.2.8 List



After entering List mode, the Mode of Power Measurement will change to LIST mode in red font, indicating that it has entered LIST mode but not yet run, and then automatically reads the parameters stored in the device.

Parameter description:

Preview: After parameter configuration is completed, a preview waveform will be automatically generated here.

List NO.: The number of the List file stored by the device, which will automatically read the parameters stored by the device after modification.

Step Number: The total number of steps, after modification, automatically generates the setting for multi-step parameters.

Repeat: The total number of iterations.

Range: Current gear, set the maximum value of the current step parameter.

Save List: Before each execution, save the parameters to the device's storage.

Process example:

1. Select the file to be executed, such as List NO. Choose 1.
2. The number of steps you need to perform in this file, such as setting Step Number to 5 and setting the parameters for each step.
3. After completing the parameter settings, set the total number of loops, such as setting Repeat to 2.
4. Set the gear and set the maximum value of the current parameter in all steps, such as setting the Range to 10A.
5. Click "Save List" to save the parameters to the device.
6. Click ON/OFF to execute.

Operation diagram

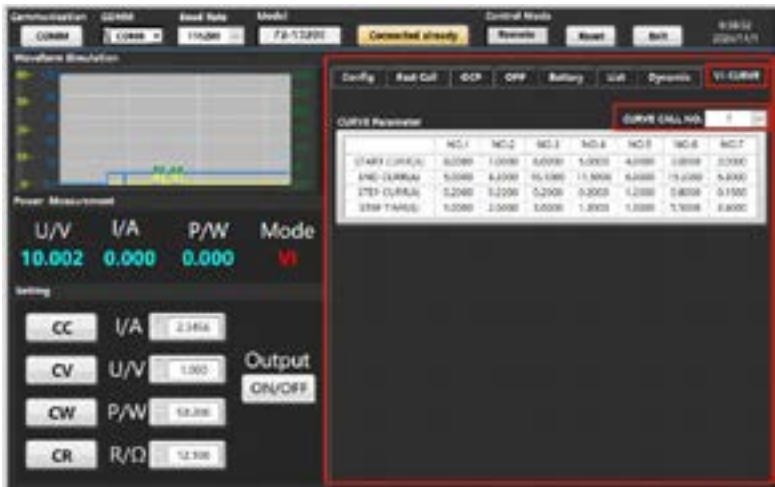


3.2.9 Dynamic



After entering Dynamic mode, the Mode of Power Measurement will change to DYN mode in red font, indicating that it has entered DYN mode but is not running. Then, the device will automatically read the stored parameters, set the corresponding parameters, select a set of parameters to execute in the Dynamic CALL NO. option box, and finally click ON/OFF on the left to run. At this time, the Mode DYN font will turn green, indicating that it is running.

3.2.10 VI-CURVE



After entering VI-CURVE mode, the Mode of Power Measurement will change to VI mode in red font, indicating that it has entered VI mode but is not running. Then, the device will automatically read the stored parameters, set the corresponding parameters, select a set of parameters to execute in the CURVE CALL NO. option box, and finally click ON/OFF on the left to run. At this time, the Mode VI font will turn green, indicating that it is running.

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